

For this assignment I chose to use DeepSeek V3 an LLM developed by Chinese company DeepSeek. For reference I will append my queries and the AIs queries separately at the bottom of this document. Initially the queries given to me by the AI were executed without issue however the final one did not run and instead threw an error as shown:

```

D:\Run\QSelect
44 JOIN Users ua ON l.link_a = ua.username near 'JOIN Users ua ON l.link_a = ua.username
45 JOIN Users ub ON l.link_b = ub.username
46 LEFT JOIN Pet p ON (ua.type = 'pet' AND p.username = ua.username) OR (ub.type = 'pet' AND p.username = ub.username)
47 LEFT JOIN Person per ON (ua.type IN ('person', 'admin') AND per.username = ua.username) OR (ub.type IN ('person', 'admin') AND per.username = ub.username)
48 SET l.pet_id = p.pet_id, l.person_id = per.person_id
49 WHERE p.pet_id IS NOT NULL AND per.person_id IS NOT NULL; 4ms

```

Upon inspection we can see that this is due to the omission of an UPDATE statement that we asked for. Now below I have inserted the SHOW CREATE results for each table.

```

CREATE TABLE `person` (
  `person_id` int NOT NULL AUTO_INCREMENT,
  `username` char(64) DEFAULT NULL,
  `nationality` char(64) DEFAULT NULL,
  PRIMARY KEY (`person_id`),
  KEY `username` (`username`),
  CONSTRAINT `person_ibfk_1` FOREIGN KEY (`username`) REFERENCES `users` (`username`)
) ENGINE=InnoDB AUTO_INCREMENT=8 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci

```

```

CREATE TABLE `pet` (
  `pet_id` int NOT NULL AUTO_INCREMENT,
  `username` char(64) DEFAULT NULL,
  `species` char(64) DEFAULT NULL,
  PRIMARY KEY (`pet_id`),
  KEY `username` (`username`),
  CONSTRAINT `pet_ibfk_1` FOREIGN KEY (`username`) REFERENCES `users` (`username`)
) ENGINE=InnoDB AUTO_INCREMENT=4 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci

```

The person and pet tables meet our instructions well, we have the correct fields, and username has been added as a foreign key. Auto-increment ensures that the primary key remains unique as long as a human doesn't manually input data for the field in an INSERT statement. Not null also ensures that no record is missing a key. Both tables meet the standards for 2NF as there are no duplicate records due to the auto-increment on the primary key, and we don't use a composite key so no partial dependency can exist. Now I am not sure whether these tables meet the standards required for 3NF as it could be said that fields like nationality or species are transitively dependent on the non-primary key attribute username. However, as pet_id and person_id map directly to username it is of no consequence in this case.

```
CREATE TABLE `links` (
  `link_id` bigint DEFAULT NULL,
  `link_a` char(64) DEFAULT NULL,
  `link_b` char(64) DEFAULT NULL,
  `creation_date` date DEFAULT NULL,
  `pet_id` int DEFAULT NULL,
  `person_id` int DEFAULT NULL,
  KEY `link_a` (`link_a`),
  KEY `link_b` (`link_b`),
  KEY `pet_id` (`pet_id`),
  KEY `person_id` (`person_id`),
  CONSTRAINT `links_ibfk_1` FOREIGN KEY (`link_a`) REFERENCES `users` (`username`),
  CONSTRAINT `links_ibfk_2` FOREIGN KEY (`link_b`) REFERENCES `users` (`username`),
  CONSTRAINT `links_ibfk_3` FOREIGN KEY (`pet_id`) REFERENCES `pet` (`pet_id`),
  CONSTRAINT `links_ibfk_4` FOREIGN KEY (`person_id`) REFERENCES `person` (`person_id`),
  CONSTRAINT `links_chk_1` CHECK ((`link_a` <> `link_b`))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
```

Here DeepSeek made a mistake with its syntax, and I could not execute its code. The fields are correct although it can be noted that auto-increment was not implemented for link_id but also of note is the fact that it was an existing field before the AI attempted to update the database and did not implement auto-increment originally. More concerningly link_id is not a primary key in this table. Therefore, this table only satisfies 1NF as you require a primary key for a 2NF table.

Appendices:

My Query:

```
CREATE TABLE Users ( username CHAR(64), password CHAR(64), email CHAR(64), type CHAR(64), login_date DATE );
```

```
ALTER TABLE Users ADD PRIMARY KEY (username);
```

```
INSERT INTO users (username, password, email, type, login_date) VALUES ('jsmith92', 'password7', 'jsmith92@gmail.com', 'person', '2025-10-27'), ('luna_the_cat', 'ivyknee', 'luna.cat@petmail.com', 'pet', '2025-10-25'), ('admin_a', 'dogsRule', 'admin.a@site.org', 'admin', '2025-10-28'), ('emma_brooks', 'emma1234', 'emma-brooks@yahoo.com', 'person', '2025-10-23'), ('rex_dog', 'treats!', 'rex@petzone.net', 'pet', '2025-10-26'), ('oliverJones', 'oj3', 'oliverjones@outlook.com', 'person', '2025-10-28'), ('bella', '256thkfgH', 'bella_bunny@outlook.com', 'pet', '2025-10-24'), ('sophie_ray_', 'rayoflightning', 'sophieray@mail.com', 'person', '2025-10-22'), ('jordan', 'password', 'jordan23@gmail.com', 'person', '2025-10-28'), ('jack7', 'secret!77', 'jack.uk@outlook.com', 'person', '2025-09-17');
```

```
CREATE TABLE Links ( link_id BIGINT, link_a CHAR(64), link_b CHAR(64), creation_date DATE );
```

```
ALTER TABLE links ADD FOREIGN KEY (link_a) REFERENCES users(username); ALTER TABLE links ADD FOREIGN KEY (link_b) REFERENCES users(username);
```

```
INSERT INTO links (link_id, link_a, link_b, creation_date) VALUES ('0', 'jsmith92', 'emma_brooks', '2025-10-21'), ('1', 'luna_the_cat', 'rex_dog', '2024-09-14'), ('2', 'oliverJones', 'sophie_ray_', '2025-06-05'), ('3', 'bella', 'luna_the_cat', '2024-09-10'), ('4', 'emma_brooks', 'bella', '2023-11-11'), ('5', 'rex_dog', 'oliverJones', '2024-12-09'), ('6', 'bella', 'jordan', '2025-02-23'), ('7', 'jack7', 'jordan', '2025-05-17'), ('8', 'jordan', 'rex_dog', '2025-10-12'), ('9', 'luna_the_cat', 'admin_a', '2025-10-27');
```

```
ALTER TABLE Users ADD CONSTRAINT CHECK (type IN ('person', 'pet', 'admin'));
```

```
ALTER TABLE links ADD CONSTRAINT CHECK (link_a <> link_b);
```

AI Query:

```
CREATE TABLE Person ( person_id INT AUTO_INCREMENT PRIMARY KEY, username CHAR(64), nationality CHAR(64), FOREIGN KEY (username) REFERENCES Users(username) );
```

```
INSERT INTO Person (username, nationality) SELECT username, CASE WHEN RAND() < 0.2 THEN 'USA' WHEN RAND() < 0.4 THEN 'UK' WHEN RAND() < 0.6 THEN 'Canada' WHEN RAND() < 0.8 THEN 'Australia' ELSE 'Germany' END as nationality FROM Users WHERE type IN ('person', 'admin');
```

```
CREATE TABLE Pet ( pet_id INT AUTO_INCREMENT PRIMARY KEY, username CHAR(64), species CHAR(64), FOREIGN KEY (username) REFERENCES Users(username) );
```

```
INSERT INTO Pet (username, species) SELECT username, CASE WHEN username = 'luna_the_cat' THEN 'Cat' WHEN username = 'rex_dog' THEN 'Dog' WHEN username = 'bella' THEN 'Rabbit' ELSE 'Unknown' END as species FROM Users WHERE type = 'pet';
```

```
ALTER TABLE Links ADD COLUMN pet_id INT; ALTER TABLE Links ADD COLUMN person_id INT;
```

```
ALTER TABLE Links ADD FOREIGN KEY (pet_id) REFERENCES Pet(pet_id); ALTER TABLE Links ADD FOREIGN KEY (person_id) REFERENCES Person(person_id);
```

```
JOIN Users ua ON l.link_a = ua.username JOIN Users ub ON l.link_b = ub.username LEFT JOIN Pet p ON (ua.type = 'pet' AND p.username = ua.username) OR (ub.type = 'pet'
```

```
AND p.username = ub.username) LEFT JOIN Person per ON (ua.type IN ('person',  
'admin') AND per.username = ua.username) OR (ub.type IN ('person', 'admin') AND  
per.username = ub.username) SET l.pet_id = p.pet_id, l.person_id = per.person_id  
WHERE p.pet_id IS NOT NULL AND per.person_id IS NOT NULL;
```